

Non-PFAS Media Lube for Hard Disk Drives (HDDs)?

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The industry-standard media lube used in HDDs, i.e., perfluoropolyether (PFPE), belongs to the broader class of per- and polyfluoroalkyl substances (PFAS). PFAS raise significant environmental and human-health concerns because of their persistence, toxicity, and potential for bioaccumulation. Increasing regulatory pressure and public scrutiny have accelerated global efforts to reduce PFAS use, creating long-term regulatory risk for the HDD industry and an urgent need for non-PFAS alternatives. To address this challenge, we evaluated polydimethylsiloxane (PDMS) as a potential non-PFAS lubricant candidate. In this work, we established a dip-coating procedure to deposit nanometer-thick lube on solid substrates. The lube thickness, uniformity, surface tension, friction, and thermal stability were systematically investigated. The results indicate that PDMS is a promising candidate, although significant challenges still remain.

Index Terms— hard disk drive (HDD), media lube, Non-PFAS, polydimethylsiloxane(PDMS)

I. INTRODUCTION

The head-disk interface (HDI) in hard disk drives (HDDs) is protected by a nanometer-thick lubricant, i.e., perfluoropolyether (PFPE), whose uniformity, bonded/mobile balance, and low surface tension govern tribological reliability. [1], [2]. PFPE belongs to the broader class of per- and polyfluoroalkyl substances (PFAS), which pose significant environmental and human-health concerns due to their persistence, toxicity, and bioaccumulation [3]. Increasing regulations and public scrutiny have accelerated global efforts to reduce PFAS usage, creating long-term regulatory risk for the HDD industry and an urgent need for non-PFAS alternatives. Polydimethylsiloxane (PDMS) has emerged as a promising non-PFAS candidate for HDD media lube. PDMS has low surface tension (~ 20 mN/m) [4] comparable to that of PFPE, positioning PDMS as the most promising fluorine-free substitutes. The friction coefficient of bulk PDMS is as low as ~ 0.138 [5], which is also comparable to PFPE. Importantly, PDMS is significantly more environmentally friendly than PFAS. While both are chemically stable, PDMS is degradable over long timescales, non-toxic, and non-bioaccumulative [6]. Moreover, PDMS is commercially available in a wide range of architectures—including variations in side chains, end groups, and molecular weights—due to its broad industrial use. Despite these advantages, a critical knowledge gap remains. While PDMS is widely studied in bulk and as micrometer-scale coatings [7], its property as a nanometer-thick film, which is essential for HDD media lubrication, is completely unexplored. To date, few studies have reported the performance, stability, or interfacial characteristics of PDMS in nanometer-scale coatings. Here, we have successfully developed a dip-coating solvent system and fabricated uniform, nanometer-thick polydimethylsiloxane (PDMS) films with hydroxyl end groups to enable bonding to the disk surface. Friction measurements, thermal baking tests, atomic force microscopy (AFM) and contact angle analyses collectively indicate that PDMS is promising non-PFAS media lube, although significant challenges still remain.

II. EXPERIMENTAL METHODS

Hydroxyl-terminated PDMS ($M_w \approx 2000$ g/mol, 65 cSt) was

dissolved in non-PFAS solvents at the desired concentration and dip-coated onto silicon wafers with native oxide at a withdrawal speed of 60 mm/min, with the desired dwell times. As-coated films contain both bonded and mobile PDMS fractions; the bonded fraction was determined after two sequential solvent rinse cycles (20 s dwell, 60 mm/min), which remove the mobile fraction. Film thicknesses were measured by spectroscopic ellipsometry (J. A. Woollam) using the native-oxide thickness fit first and the PDMS layer modeled with a Cauchy dispersion ($n \approx 1.43$) model. Surface morphology were characterized by AFM in tapping mode; static water (WCA) and hexadecane (HCA) contact angles were recorded using the sessile-drop method. Friction coefficient was measured using a ball-on-disk tribometer.

III. RESULTS AND DISCUSSION

A. Thickness vs. Solution Concentration

Fig. 1 summarizes the characterization results obtained using the established dip-coating procedure. As shown in Fig. 1, the total PDMS thickness increases with solution concentration, as expected. Film thickness was determined by ellipsometry using protocols previously established in our laboratory. The bonded thickness initially increases with solution concentration and then levels off, consistent with a solution adsorption mechanism reported in a prior study [1]. At the thickness relevant for HDD applications (~ 1 nm), the bonded thickness is approximately 0.5 nm, which is comparable to that of PFPE.

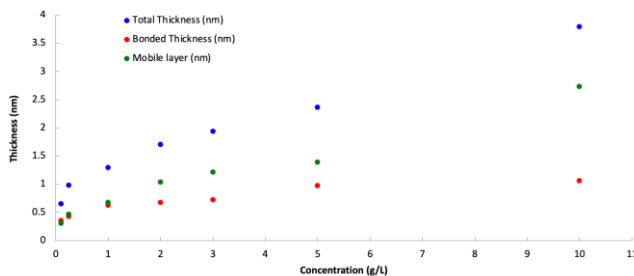


Fig. 1. Total and bonded/mobile PDMS thickness as a function of solution concentration (dwell time 20 s, withdrawal speed 60 mm/min).

B. Surface Morphology

Fig. 2 shows AFM images and R_q values for ~ 1 nm PDMS.

The AFM image clearly demonstrates the excellent uniformity, with a root-mean-square roughness (R_q) comparable to that of the bare Si wafer substrate.

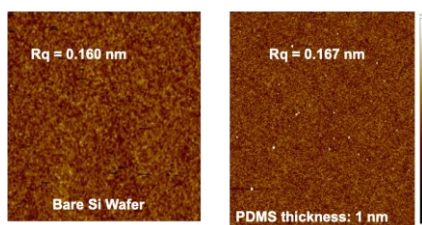


Fig. 2. AFM characterization of PDMS/Si wafer with Bare Si Wafer reference.

C. Wettability

Fig. 3a shows static water (WCA) and hexadecane (HCA) contact angles on as-coated PDMS films on Si wafer. The PDMS/Si exhibits a higher water contact angle (WCA) and a lower hexadecane contact angle (HCA) than PFPE/Si. Surface energy components of PFPE/Si and PDMS/Si, calculated using the Fowkes model from water and hexadecane contact-angle measurements [8], are shown in Fig. 3b. The results indicate that the total surface energy of PDMS/Si is even lower than that of PFPE/Si. However, the dispersive component of PDMS/Si is higher than that of PFPE/Si, suggesting lower resistance to airborne hydrocarbon. This could be a concern for long-term HDI reliability.

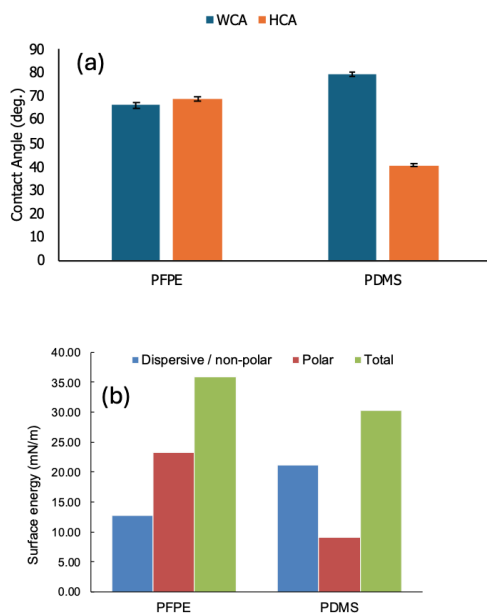


Fig. 3. (a): Static water (WCA) and hexadecane (HCA) contact angles of ~ 1 nm PDMS on Si wafer. (b): Surface energy of ~ 1 nm PDMS on Si wafer.

D. Friction & Thermal Stability

As shown in Fig. 4a, the friction coefficient of the ~ 1 nm PDMS/Si is higher than its bulk value, as expected for nanometer-thick lubricant films, and is slightly higher than that of ~ 1 nm PFPE/Si when considering the size of the error bars. Thermal stability was evaluated using a baking test at 150°C for 2 hours (Fig. 4b). The lube loss for ~ 1 nm PDMS is approximately 16%, compared to $\sim 8\%$ for PFPE. This higher loss can be attributed to the broader molecular weight distribution (MWD) of PDMS, which typically exhibits

polydispersity indices (PDI) ranging from ~ 1.7 to ~ 3.5 [9], whereas a perfectly monodisperse polymer has $\text{PDI} = 1$. When the MWD is broad, the low-molecular-weight fraction in nanometer-thick lubricant films can evaporate under thermal stress, leading to increased lubricant loss. In contrast, PFPE lubricants undergo supercritical fluid extraction (SFE) to achieve extremely narrow MWDs ($\text{PDI} \approx 1.01\text{--}1.03$).

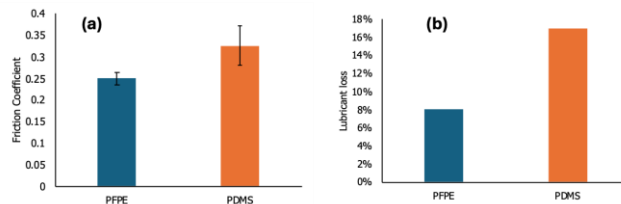


Fig. 4. (a): Friction coefficient of ~ 1 nm PDMS/Si and ~ 1 nm PFPE/Si. (b): Lubricant loss of ~ 1 nm PDMS/Si and ~ 1 nm PFPE/Si after baking test (150°C for 2 hours).

IV. CONCLUSION

A repeatable dip-coating process for hydroxyl-terminated PDMS nanofilms on native-oxide Si wafer has been established, producing films from sub-nanometer to several nanometers thick. Solution concentration is the dominant control of total and bonded thickness, while dwell time provides secondary tuning. AFM confirms continuous, smooth films with no dewetting. Water and hexadecane contact angles measurement show that the PDMS/Si has lower surface tension than PFPE/Si. These results establish PDMS nanofilms as a promising non-PFSA candidate for media lube. However, lower resistance to airborne hydrocarbon, higher friction and lower thermal stability remains significant challenges.

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